

9/21/2026

Last time:

Thm (Existence of prime factorizations in $\mathbb{Z}[\sqrt{d}]$) Assume $d \in \mathbb{Z}$

is not a square. Every nonzero, nonunit element can be

written as a (finite) product of primes.

Examples of nonunique factorizations

1. $\mathbb{Z}[\sqrt{-3}]$ ($d = -3$): $4 = 2 \cdot 2 = (1 + \sqrt{-3})(1 - \sqrt{-3})$.

2. $\mathbb{Z}[\sqrt{-5}]$ ($d = -5$): $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$.

3. $\mathbb{Z}[\sqrt{5}]$ ($d = 5$): $4 = 2 \cdot 2 = (3 + \sqrt{5})(3 - \sqrt{5})$.

① 2 is prime

Let $\alpha \mid 2$. Then $|N(\alpha)| \mid |N(2)| = 2^2 = 4$. So

$$|N(\alpha)| = 1, 2, 4.$$

① $|N(\alpha)| = 1$: α is a unit

② $|N(\alpha)| = 4$: $|N(\alpha)| = |N(2)|$. So α is an associate of 2.

③ $|N(\alpha)| = 2$: Assume $\alpha = x + y\sqrt{5}$. Then

$$|N(\alpha)| = |x^2 - 5y^2| = 2.$$

$$\text{So } x^2 - 5y^2 = \pm 2.$$

Mod 5: $x^2 \equiv \pm 2 \pmod{5}$. Hence, $2 \not\equiv \pm 2 \pmod{5}$ or $-2 \not\equiv \pm 2 \pmod{5}$.

Since $5 \equiv 5 \pmod{8}$,

$$\left(\frac{2}{5}\right) = -1, \quad \left(\frac{-2}{5}\right) = \underbrace{\left(\frac{-1}{5}\right)}_{1} \underbrace{\left(\frac{2}{5}\right)}_{-1} = -1.$$

So $\pm 2 \nmid 5$. A contradiction.

Thus, α is either a unit or an associate of 2.

So 2 is prime.

② $\alpha = 3 \pm \sqrt{5}$ are prime

Let $\beta \mid \alpha$. Then $|N(\beta)| \mid |N(\alpha)| = 4$. So $|N(\beta)| = 1, 2, 4$.

By the same argument, β is either a unit or an

associate of α . Hence, α is prime.

Recall:

Division algorithm $\Rightarrow \mathbb{Z}$ is a PID

$\stackrel{\text{gcd}}{\Rightarrow}$ Bézout's lemma

$\Rightarrow$ Euclid's lemma

$\Rightarrow$ Uniqueness.

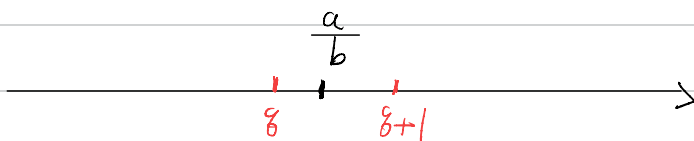
Division algorithm in $\mathbb{Z}$: Given $a, b \in \mathbb{Z}$ with $b \neq 0$,

there are unique $q, r \in \mathbb{Z}$ i.e.

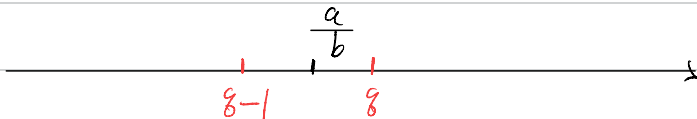
$$a = qb + r, \quad 0 \leq r < |b|.$$

pf idea: Take

$$g = \begin{cases} \text{greatest integer} \leq \frac{a}{b} & , \text{ if } b > 0, \\ \text{least integer} \geq \frac{a}{b} & , \text{ if } b < 0. \end{cases}$$



$$b > 0$$



$$b < 0$$

This time, take $g = \text{integer closest to } \frac{a}{b}$ instead,

then $\left| \frac{a}{b} - g \right| \leq \frac{1}{2}$. If $r = a - gb$, then

$$|r| = |a - gb| = |b| \cdot \underbrace{\left| \frac{a}{b} - g \right|}_{\leq \frac{1}{2}} \leq \frac{1}{2} |b|.$$

What did we lose?

① r may be negative.

② g, r are no longer unique when $\frac{a}{b} = n + \frac{1}{2}$ for

some $n \in \mathbb{Z}$.

Division algorithm in $\mathbb{Z}[\sqrt{d}]$ for $d = -1, \pm 2, 3$.

Thm (DA in $\mathbb{Z}[\sqrt{d}]$) Given $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$ with $\beta \neq 0$,

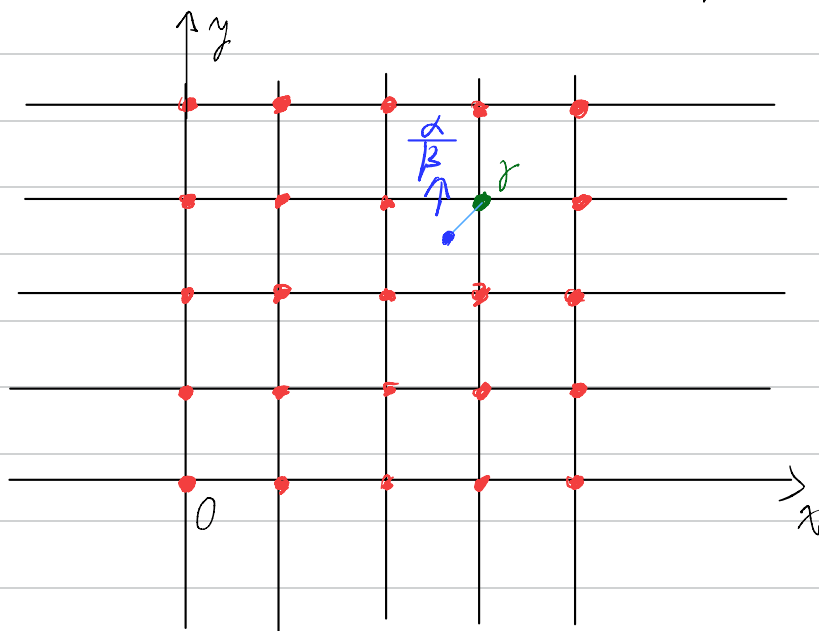
there are $r, s \in \mathbb{Z}[\sqrt{d}]$ s.t.

$$\alpha = \beta r + s, \quad |N(s)| < |N(\beta)|.$$

Idea: Take Gaussian integers $a+bi$ ($a, b \in \mathbb{Z}$)

corresponding to $d = -1$.

$a+bi$ $\longleftrightarrow$ lattice point (a, b)



$x_0 + iy_0$
||

Take $\frac{\alpha}{\beta} = x + iy \in \mathbb{C}$ and r to be an integer

closest to $\frac{\alpha}{\beta}$. Then $N(\frac{\alpha}{\beta} - r) = \underbrace{|x - x_0|^2}_{\leq \frac{1}{2}} + \underbrace{|y - y_0|^2}_{\leq \frac{1}{2}} \leq \frac{1}{2}$.

If $\delta = \alpha - \beta r$, then

$$N(\delta) = N(\beta \cdot (\frac{\alpha}{\beta} - r)) = N(\beta) \underbrace{N(\frac{\alpha}{\beta} - r)}_{\leq \frac{1}{2}} \leq \frac{1}{2} N(\beta).$$

Will prove the general case next time.